

Benchmarking Deep Learning Architectures for Automated Knee MRI Classification

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Abstract. Knee MRI interpretation is time-intensive and prone to variability, motivating automated decision support for ACL and meniscal tear classification. We benchmark three deep learning architectures on the MRNet dataset (1,370 examinations; three imaging planes; binary labels for abnormality, ACL tear, and meniscal tear): a ResNet18 baseline, an attention-pooling ResNet18 with multi-plane fusion, and a Swin Transformer fusion model. Performance is evaluated using ROC-AUC, PR-AUC, sensitivity, and specificity; Grad-CAM is applied for spatial interpretability. The baseline ResNet18 fusion model achieved the strongest aggregate performance, with ROC-AUC 0.898, sensitivity 90.5%, and specificity 75.0%. The Swin Transformer fusion model performed best for abnormality classification (ROC-AUC 0.928). These results support the feasibility of deep learning as a triage tool for ACL and meniscal pathology, pending external validation and prospective clinical evaluation.

Keywords: Knee MRI, deep learning, vision transformer, convolutional neural network, Grad-CAM, musculoskeletal imaging

1 Introduction and clinical motivation

Knee magnetic resonance imaging (MRI) is the preferred method for diagnosing intra-articular injury, yet each examination requires systematic review across three anatomical planes and may contain several hundred slices, placing considerable demands on musculoskeletal radiologists [1]. Anterior cruciate ligament (ACL) reconstruction rates have risen from 40.6 to 51.2 procedures per 100,000 population over 17 years, with disproportionate growth among females and younger cohorts [2]; meniscal tears are equally prevalent, with 52.8% of confirmed cases requiring arthroscopic surgery and 30.3% presenting with concomitant ligamentous injury [3], conditions that co-occur in a substantial proportion of ACL-injured knees [4].

This burden motivates automated triage support for knee MRI reporting. Bien et al. [1] demonstrated that a convolutional neural network (CNN) trained on the MRNet dataset could match or exceed unassisted radiologists on abnormality and ACL tear detection, and that model predictions significantly improved diagnostic specificity. The present study extends this through a systematic multi-methodology evaluation: we benchmark three deep learning architectures: a ResNet18 baseline, an attention-pooling ResNet18 with multi-plane fusion, and a Swin Transformer fusion model. A

Wasserstein GAN with Gradient Penalty (WGAN-GP) was additionally applied as a data augmentation strategy to mitigate class imbalance. We also apply unsupervised clustering to residual ambiguous cases and deploy Gradient-weighted Class Activation Mapping (Grad-CAM) to interrogate each model's spatial predictions. All experiments use the MRNet dataset [1], comprising 1,370 knee MRI examinations with binary labels for abnormality, ACL tear, and meniscal tear.

2 Related Work

The MRNet architecture [1] established the benchmark for deep learning-based knee MRI classification using separate per-plane CNNs for the axial, coronal, and sagittal series, combined via logistic regression. MRNet achieved area under the curve (AUC) values of 0.937, 0.965, and 0.847 for abnormality, ACL tear, and meniscal tear detection, respectively, matching or exceeding unassisted radiologists on two of three tasks and significantly improving clinician specificity when predictions were provided.

Subsequent work confirmed human-level CNN performance on individual knee MRI tasks; Minamoto et al. [5] achieved sensitivity of 91.0%, specificity of 86.0%, and accuracy of 88.5% for ACL tear detection using a sagittal CNN. Transfer learning from ImageNet-pretrained weights has consistently improved performance in low-data medical imaging settings, motivating pretrained backbones in our baseline and Swin Transformer models [6]. Class imbalance in MRNet, where confirmed pathology is the minority class, is addressed via GAN-based augmentation; Liu et al. [7] demonstrated that synthesised training images of sufficient quality yield substantial accuracy improvements when labelled data are limited.

Swin-based vision transformer architectures offer hierarchical self-attention across three-dimensional volumes and have achieved competitive performance across medical imaging benchmarks [8]; applying the Swin Transformer fusion model to MRNet treats each examination volumetrically, capturing through-plane context unavailable to per-slice approaches. Grad-CAM [9], the most widely used explainability method in radiological imaging [10], generates class-discriminative activation maps validated against lesion sites in CT-based tasks [11].

3 Dataset and Preprocessing

3.1 Dataset

The dataset is highly imbalanced with 80.6% of samples being labelled abnormal vs 19.4% normal, with 37.1% meniscal tears and 23.3% ACL tears. This also leaves us with 20.2% abnormal scans with an unknown injury. The validation set was used as a holdout test set, containing 119 samples, and the training set was split into a train and validation set using an 85:15 split (960/170 samples).

Custom PyTorch Dataset and DataLoader were implemented using the PyTorch API to load the MRNet dataset. The dataset loads MRI volumes based on the specified split and imaging plane, and pairs each scan with its corresponding label.

3.2 Preprocessing

Model-specific preprocessing is applied, including intensity normalisation, slice-wise data augmentation, and conversion to tensors. MRI volumes are not resized to a fixed number of slices, instead, the full volume is used as provided in the dataset. Intensities are normalised either by scaling to $[0, 1]$ or using min-max scaling depending on the model setup. During training, data augmentation is applied per slice using random rotations, small affine translations, and horizontal flips to improve robustness.

3.3 Exploratory Clustering

Various clustering techniques such as principal component analysis (PCA) and KMeans were used to explore whether the abnormal scans could be clustered into more detailed groups. No technique was able to group them based on physiological issue; rather, the clusters all contained equal parts of each abnormal issue (ACL, meniscal tear and other abnormal), indicating the clustering was based on some image artefact rather than physiological property. Results can be found in Appendix C.

4 Model architecture/training

Figure 5 in Appendix B illustrates the full pipeline for this MRNet analysis. The baseline model was implemented as a CNN for knee MRI classification. Each MRI exam was processed using one anatomical plane at a time, including axial, coronal, or sagittal scans. Since each MRI scan contains multiple 2D slices but only one binary label, the model first extracted features from each slice and then combined them into one representation for final classification.

A pretrained ResNet18 was used as the backbone feature extractor. The final classification layer of ResNet18 was removed, so each MRI slice was converted into a feature vector rather than being classified independently. The feature vectors from the same MRI exam were then combined using max pooling across the slice dimension. The reason for using max pooling is that ACL and meniscus abnormalities may only be clearly visible in a small number of slices. Therefore, max pooling helps retain the strongest abnormality-related signal across the full MRI exam.

After slice aggregation, the resulting feature vector was passed through a dropout layer and a fully connected layer to produce one binary output logit. The dropout layer was included to reduce overfitting before the final classification layer. The fully connected layer then mapped the aggregated MRI representation to a single output value for the binary classification task. To improve upon the max-pooling baseline, the slice aggregation mechanism was replaced with a learned attention-pooling module. Rather than selecting the single strongest slice response, the attention module assigns a

scalar weight to each slice based on its learned diagnostic relevance. Each slice feature vector was passed through a small two-layer network producing a scalar score, which was normalised across all slices using softmax. The final exam-level representation was computed as the weighted sum of all slice feature vectors, allowing the model to focus on diagnostically informative slices while suppressing uninformative ones.

The attention-based model retained the same ResNet18 backbone and classification head, with max pooling replaced by this learned aggregation. A two-stage training protocol was applied to prevent overfitting. The backbone was first frozen for ten epochs while only the attention module and classifier were trained, then the full network was fine-tuned end-to-end using differential learning rates.

To further improve performance, the ResNet18 backbone was replaced with a Swin Transformer (Swin-Tiny, pretrained on ImageNet). Unlike convolutional networks, the Swin Transformer computes self-attention within local windows and uses a shifted window mechanism across layers to capture both fine-grained local features and broader spatial context, making it well-suited to detecting subtle anatomical abnormalities in MRI slices. The backbone produced a 768-dimensional feature vector per slice, aggregated using the same attention pooling mechanism. Multi-plane fusion was performed by training one model per anatomical plane and combining the three output probabilities using a weighted sum, with weights optimised on the validation set using the Nelder-Mead algorithm to maximise AUC.

5 Experimental design/metrics

5.1 Experiments

As mentioned, the baseline model was selected as ResNet18 with max pooling, a pre-trained model, and using the trained weights. A dropout rate of 0.3 was applied before linear mapping to output logit. The model was also trained using the Adam optimiser with a learning rate of 1×10^{-4} and weight decay of 1×10^{-4} . The class imbalance was addressed by using binary cross-entropy loss with the positive class weighted by the negative-to-positive ratio. The learning rate used a reduce learning rate on plateau by a factor of 0.3 and patience of 2 epochs. The training was run for 20 epochs with early stopping applied after 5 epochs without improvement.

To modify the ResNet18 model, max pooling from the baseline was replaced with an attention mechanism. Model 2 trains 9 separate models for each injury across the 3 planes, combining the predictions at the end. Model 2 used a linear layer followed by a Tanh activation and finally a second linear layer, then a softmax to compute the weighted sum of the slice features. A dropout rate of 0.5 was also applied and trained using binary cross-entropy loss. The encoder was trained using a learning rate of 1×10^{-5} and the gate and classifier using 1×10^{-4} , with the scheduler dropping the learning rate by a factor of 0.5 with a patience of 3. Model 2 was trained for a maximum of 50 epochs with early stopping at 10 epochs.

Following on from the ResNet18 with an attention mechanism, the Swin Transformer was adopted. Instead of replacing only the pooling, the ResNet18

backbone was replaced with a Swin Transformer. Rather than filters sliding across an image, the image is divided into patches in a 4x4 pixel layout. The image is resized in a 224x224 layout, giving a total of 56 patches per side, for a total of 3136 patches. The token structure of the transformer, each patch is treated as a token. Model 3: the first 10 epochs were trained at 1×10^{-3} , after the backbone was unfrozen, the learning rate was dropped to 1×10^{-5} for the trunk and to 1×10^{-4} for the gate and classifier. Binary cross-entropy loss was also used with a reduction on plateau of a factor of 0.5 and patience 3. Training lasted a maximum of 50 epochs, and early stopping had a patience of 10 epochs.

5.2 Other Experiments

A Wasserstein GAN with Gradient Penalty (WGAN-GP) was implemented. The model examined real images to produce synthetic images, aiming to determine if the synthetically produced scans could be passed as real ones. A standard GAN was used as a baseline, achieving a fool rate of $\sim 20\%$. Switching the model to a WGAN-GP improved the fool rate to $\sim 30\%$. A conditional component was added to classify real and synthetic images, with the critic achieving an accuracy of 75% for real and 25% for synthetic. Their models were trained on each plane (Sagittal, Coronal, Axial). Experimentation was conducted by modelling each injury for each plane, which shrank their respective dataset, yielding far worse results than a per-plane model.

Distillation with No Label v3 (DINOv3) is a vision transformer like Swin Transformer. DINOv3 is a self-supervised learning approach using pretrained weights. The model results did not perform as strongly as the Swin Transformer, as DINO was trained on natural images rather than MRI scans.

5.3 ROC-AUC/PR-AUC

The dataset classes are imbalanced, with far more abnormal cases than normal cases. Abnormal cases also have issues, since 20.2% of cases are not annotated for a specific injury excluding ACL and meniscus. Receiver operating characteristic AUC (ROC-AUC) is used as the primary model selection criterion, as it measures how well the model ranks positive to negative classes. This metric was used in the original MRNet paper [1], making direct comparisons possible. Precision-Recall AUC (PR-AUC) was also reported due to the imbalance in the dataset. PR-AUC is a vital metric, as it captures performance on the minority class. This prevents the model from simply predicting the majority classes and reporting strong results.

5.4 Other Metrics

All models deal with the classification of classes, and sensitivity is a vital metric of the model being able to correctly identify true tears. High sensitivity is key when dealing with medical analysis. Specificity, the proportion of healthy knees correctly identified as negative. The F1 metric was useful due to the imbalance of the dataset; the metric is

not inflated by an increased number of true negatives, providing a more informative score of the model's overall performance.

6 Results/error analysis

6.1 Classification Performance

Table 2 summarises held-out validation performance for the baseline ResNet18 fusion model, enhanced ResNet18 fusion model and Swin Transformer fusion model, aggregated across the three MRNet tasks. ROC-AUC was used as the primary discrimination metric, with PR-AUC included because ACL and meniscal tear labels are less prevalent than the broad abnormality label. Sensitivity, specificity and accuracy were calculated using a probability threshold of 0.5. For all three model families, sagittal, coronal and axial predictions were combined using equal probability fusion.

Table 2. Validation performance of candidate models on the MRNet validation set.

Aggregate held-out validation performance across MRNet outcomes

Equal sagittal/coronal/axial probability fusion; pooled abnormality, ACL tear, and meniscal tear task-level predictions; threshold metrics use $P \geq 0.5$.

Model	N (positive)	ROC-AUC	PR-AUC	Sensitivity	Specificity	Accuracy	F1
Baseline ResNet18 fusion	357 (201)	0.898	0.910	0.905	0.750	0.838	0.863
Enhanced ResNet18 fusion	357 (201)	0.859	0.890	0.816	0.763	0.793	0.816
SwinViT fusion	357 (201)	0.887	0.907	0.881	0.731	0.815	0.843

Higher values are better. Bold indicates the best aggregate value in each metric column.

N (positive) reports pooled task-level predictions, not unique patients; fusion weights are equal and not tuned on held-out labels.

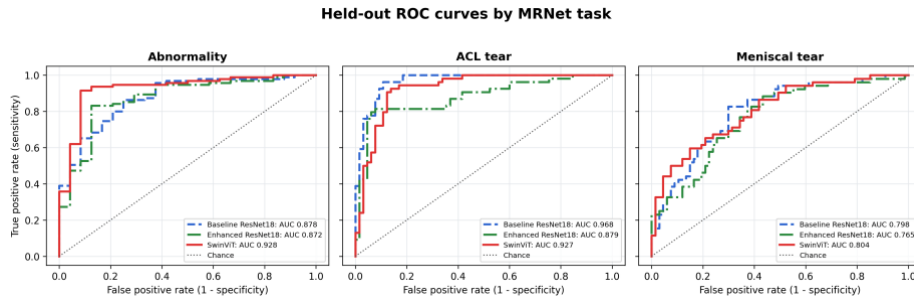


Figure 3. ROC curves for abnormality, ACL tear and meniscal tear classification.

Overall, the baseline ResNet18 fusion model achieved the strongest aggregate performance across pooled task-level predictions, with ROC-AUC 0.898, PR-AUC 0.910, sensitivity 90.5%, specificity 75.0%, accuracy 83.8% and F1 0.863. Swin Transformer fusion performed similarly overall, with ROC-AUC 0.887, PR-AUC

0.907, sensitivity 88.1%, specificity 73.1% and accuracy 81.5%. The enhanced ResNet18 fusion model had the highest aggregate specificity at 76.3%, but lower sensitivity and discrimination than the baseline model.

At task level, Swin Transformer fusion showed the strongest abnormality classification, with ROC-AUC 0.928 and PR-AUC 0.978. Baseline ResNet18 fusion performed best for ACL tear classification, with ROC-AUC 0.968 and PR-AUC 0.956 and also gave the strongest aggregate pooled results. Meniscal tear classification remained the most challenging task across all models, with lower ROC-AUCs and weaker specificity than abnormality or ACL tear classification.

These results suggest that, under the final held-out evaluation protocol, the simpler ResNet18 fusion model was the most reliable overall candidate. The Swin Transformer model remained competitive and performed particularly well for abnormality detection but did not consistently outperform the ResNet baseline.

6.2 Qualitative error analysis and explainability

Grad-CAM was applied to representative validation cases from the ACL/sagittal (ResNet) model to assess whether predictions were driven by anatomically plausible regions. Maps were generated for the positive ACL class and examples of true positive (TP), false positive (FP), true negative (TN), and false negative (FN) predictions are shown in Figure 4.

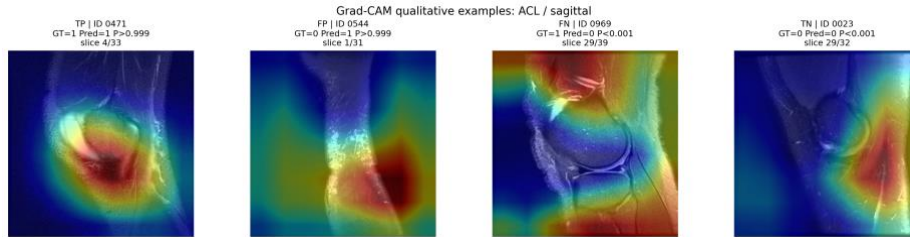


Figure 4. Grad-CAM examples for ACL/sagittal classification. GT is the Ground Truth label. Pred is the models Predicted label. P is the probability assigned by the model.

The TP case showed activation near the expected ACL region, suggesting clinically plausible attention. In contrast, the FP displayed more diffuse activation extending into peripheral regions, indicating possible reliance on non-specific image features. The FN showed little activation around the ACL despite a positive label, suggesting that the single plane model may miss subtle findings. The TN also exhibited weak peripheral activation, highlighting that Grad-CAM in negative cases reflects positive class evidence rather than normal anatomy.

Overall, Grad-CAM provided useful qualitative insight but revealed limitations in model attention, particularly in error cases where activation was diffuse or peripheral. These findings supported further improvements such as multi-plane modelling, enhanced preprocessing, and attention-based approaches. Grad-CAM was used primarily as an error analysis tool rather than evidence of radiologist-like reasoning.

7 Discussion/limitations/bias/generalization

The baseline ResNet18 fusion model achieved the strongest aggregate ROC-AUC of 0.898, outperforming the Swin Transformer fusion model overall (aggregate ROC-AUC 0.887). The Swin Transformer fusion model was strongest for abnormality classification (ROC-AUC 0.928). Per-task ROC-AUC values for the Swin Transformer fusion model were 0.928, 0.927 and 0.804 for abnormality, ACL, and meniscal tear detection, respectively. Although these results remain broadly comparable to the original MRNet benchmark of 0.937, 0.965, and 0.847 reported by Bien et al. [1], performance was lower across most tasks. The strong ACL detection performance is consistent with Minamoto et al. [5], who demonstrated that CNN-based approaches can approach human-level performance on individual knee MRI tasks, while the application of a Swin Transformer architecture aligns with Ali et al. [8], whose scoping review identified Swin-based models as among the strongest architectures across medical imaging benchmarks.

Several limitations qualify these findings. The MRNet dataset [1] originates from a single institution, limiting the capture of variability in acquisition protocols and patient demographics; the dataset's skew towards male patients (801 males, 569 females) is particularly concerning given epidemiological evidence that ACL reconstruction rates have risen disproportionately among females [2] and that meniscal tears carry a substantial surgical burden across mixed-sex populations [3]. The binary labelling scheme also fails to reflect the clinical complexity identified by Kilcoyne et al. [4], who demonstrated that meniscal tears co-occur with ACL injuries in a substantial proportion of cases. Class imbalance, only partially mitigated by the GAN augmentation approach of Liu et al. [7], and the clearly defined diagnostic categories may have further simplified the classification task and inflated the reported metrics. External validation on larger multi-centre datasets with diverse imaging protocols and patient demographics is required before the approach can be considered suitable for clinical deployment.

8 Conclusion/future work

Three deep learning architectures were benchmarked for automated knee MRI classification on the MRNet dataset, with WGAN-GP augmentation applied to address class imbalance. The ResNet18 baseline achieved the strongest aggregate performance (ROC-AUC 0.898, sensitivity 90.5%, specificity 75.0%), while the Swin Transformer fusion model led on abnormality classification (ROC-AUC 0.928). These results support the feasibility of deep learning for ACL and meniscal triage, pending external validation, probability calibration, and prospective clinical evaluation.

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Appendix A:

To test for statistical significance between the models, DeLong's test was used. Comparing the AUC-ROC values using a paired non-parametric test, accounting for correlations between models. All pairwise comparisons were conducted across each injury and plane type. To control the risk of false positives, a Bonferroni Correction ($\alpha=0.0056, 0.05/9$) was applied.

Table 1. Pairwise comparison of model AUCs across MRNet conditions.

Condition	Comparison	AUC A	AUC B	Δ AUC	p-value
ACL	Baseline vs ResNet18	0.968	0.879	-0.089	0.005 *
	Baseline vs SwinViT	0.968	0.927	-0.041	0.042
	ResNet18 vs SwinViT	0.879	0.927	+0.048	0.090
Meniscus	Baseline vs ResNet18	0.798	0.765	-0.033	0.429
	Baseline vs SwinViT	0.798	0.804	+0.006	0.849
	ResNet18 vs SwinViT	0.765	0.804	+0.039	0.268
Abnormal	Baseline vs ResNet18	0.878	0.872	-0.006	0.921
	Baseline vs SwinViT	0.878	0.928	+0.050	0.237
	ResNet18 vs SwinViT	0.872	0.928	+0.056	0.142

Appendix B:

MRNet modelling pipeline

Three-plane MRI exams are processed per plane, fused at probability level, and evaluated on held-out validation data

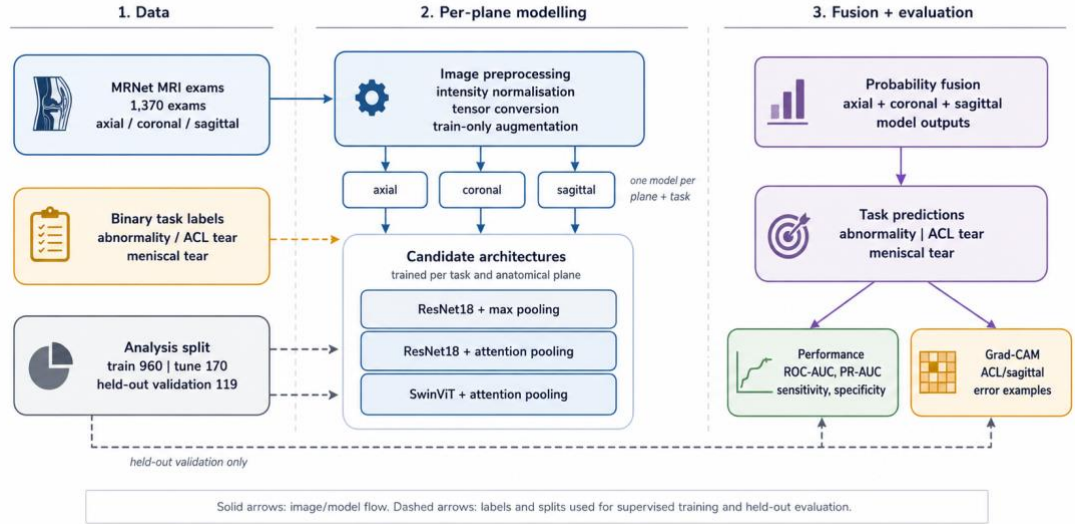


Figure 5. MRNet modelling pipeline. MRNet examinations were preprocessed, modelled separately by anatomical plane and task, fused at probability level across axial, coronal and sagittal outputs, and evaluated on held-out validation data using discrimination, threshold-based metrics and Grad-CAM error analysis.

Appendix C:

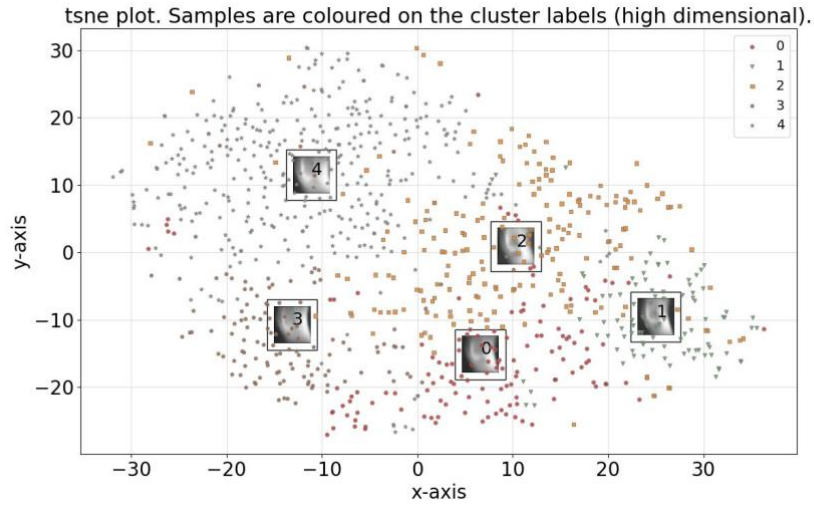


Table 2. Diagnosis rates by patient cluster.

Cluster	Total Cases	ACL Rate	Meniscus Rate	ACL or Meniscus Rate	Pure Abnormal Rate
0	203	28.1%	39.9%	51.2%	48.8%
1	183	19.1%	43.7%	51.9%	48.1%
2	166	21.7%	49.4%	55.4%	44.6%
3	135	26.7%	35.6%	48.9%	51.1%
4	225	19.6%	47.1%	54.7%	45.3%